

SECTION AAnswer **all** questions.

1. State why chlorine is used in water treatment. [1]

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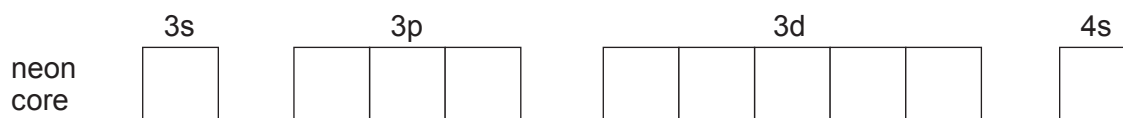
2. State the pattern in electronegativity across Period 3 in the Periodic Table. [1]

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3. Name the element in Period 3 that has a half-filled set of *p*-orbitals. [1]

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4. By inserting arrows to represent electrons, complete the electronic structure of a vanadium atom. [1]



5. Magnesium nitrate decomposes when heated to form magnesium oxide, nitrogen dioxide and oxygen.

Balance the equation for this reaction. [1]



- (iii) Use your answers to parts (i) and (ii) to show that the value of x in $\text{Na}_2\text{CO}_3 \cdot x\text{NaHCO}_3 \cdot y\text{H}_2\text{O}$ is 1.

[1]

- (iv) The relative formula mass of $\text{Na}_2\text{CO}_3 \cdot x\text{NaHCO}_3 \cdot y\text{H}_2\text{O}$ is 226. Calculate the value of y .

[2]

 $y = \dots\dots\dots$ **END OF PAPER**

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